

DAOD 3026-1, Radio Frequency Safety Program

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1. Introduction

Date of Issue: 2012-05-04

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Application: This DAOD is a directive that applies to employees of the Department of National Defence (DND employees) and an order that applies to officers and non-commissioned members of the Canadian Armed Forces (CAF members).

Supersession: CFAO 34-51, Radio Frequency Radiation Safety; CANFORGEN 121/12, DND/CF Radio Frequency Safety

Approval Authority: Assistant Deputy Minister (Materiel) (ADM(Mat))

Enquiries: Quality Engineering Test Establishment (QETE) Superintendent

2. Definitions

ammunition (munition) - A device charged with explosives, propellants, pyrotechnics, initiating composition or nuclear, biological or chemical material, for use in military operations, and includes a non-charged or inert replica of such a device. (Defence Terminology Bank record number 15923)

emission (émission) - The phenomenon by which energy emanates from a source in the form of waves or particles. (Defence Terminology Bank record number 7989)

explosives (explosif) - Any thing that is made, manufactured or used to produce an explosion or a detonation or pyrotechnic effect, and includes any thing prescribed to be an explosive by the regulations made under the Explosives Act, but does not include gases, organic peroxides or any thing prescribed not to be an explosive by the regulations made under that Act. (Section 2 of the Explosives Act)

exposure (exposition) - The amount of absorbed radiation as determined by the level of emission and the time duration of that emission. (Defence Terminology Bank record number 48285)

hazardous occurrence (situation comportant des risques) - Any event or series of events that results, or has the potential to result, in injury, illness or death, or damage to materiel, real property, immovables or the environment, regardless of the severity of the injury, illness or damage. Note: DND and CAF policies and instruction set out notification, investigation and reporting requirements for all HOs. (Defence Terminology Bank record number 694257)

radio frequency (radiofréquence) - The frequency of a periodic radio wave or of the corresponding electrical oscillation. (Defence Terminology Bank record number 7982)

radio frequency device (dispositif de radiofréquence) - A device that generates or utilizes radio frequency energy. (Defence Terminology Bank record number 48268)

radio frequency radiation (rayonnement de radiofréquence) - Energy propagating through space in form of electromagnetic waves, which: are comprised of electric and magnetic fields oscillating at frequencies in the range of 3 kHz to 300 GHz; and do not contain enough energy to remove tightly bound electrons from their orbits around atoms. (Defence Terminology Bank record number 48269)

secondary x-ray (rayon X secondaire) - Unintended X-ray radiation emitted from a radio frequency device containing an internal high voltage source exceeding 15 kilovolts. (Defence Terminology Bank record number 693843)

3. Abbreviations

CF H Svcs Gp - Canadian Forces Health Services Group

COS(Mat) - Chief of Staff (Materiel)

CO - commanding officer

comd - commander

HERF - hazards of electromagnetic radiation to fuel

HERO - hazards of electromagnetic radiation to ordnance

HERP - hazards of electromagnetic radiation to personnel

HO - hazardous occurrence

L1 - level one advisor

OHS - occupational health and safety

RF - radio frequency

RFS - radio frequency safety

RFSB - Radio Frequency Safety Board

RFSO - radio frequency safety officer

RFSOD - Radio Frequency Safety Orders and Directives

RFSP - Radio Frequency Safety Program

TA - technical authority

4. Overview

Context

4.1 This DAOD must be read in conjunction with: DAOD 3026-0, Radio Frequency Safety; and C-55-040-001/TS 002, Radio Frequency Safety Standards and Requirements.

Purpose

4.2 The purpose of the RFSP is to: provide the framework upon which L1s, managers of a DND organization, Comds of a command or formation and COs of a unit formulate and implement their own RFS orders, directives and guidance; prevent incidents, injuries and losses that may occur as a result of RF radiation exposures; and address RF radiation hazards to personnel, fuel, ammunition, explosives and other materiel, as well as hazards resulting from secondary ionizing radiation emitted by RF devices utilizing high voltages, including: HERP, HERF and HERO; and hazards of secondary X-rays. Note: Policy direction in respect of secondary X-rays is also provided by the Director Nuclear Safety.

RFSP Operational Structure

4.3 The RFSP operates within the following three-tier structure: the program authority tier: the COS(Mat), Director General Land Equipment Program Management, QETE Superintendent and RFSP manager; the TA tier: the QETE Superintendent and RFS TA; and the operational tier: managers of a DND organization, Comds of a command or formation and COs of a unit that hold or use RF devices. Note: For the respective responsibilities of the office holders in each tier of the RFSP operational structure, see the Responsibility Table.

5. Radio Frequency Safety Program Compliance

Federal Standards

5.1 The DND and the CAF must ensure that the RFSP: complies with the requirements of: Part II of the Canada Labour Code; the Canada Occupational Health and Safety Regulations; and exposure guidelines set out in the Limits of Human Exposure to Radiofrequency Electromagnetic Fields in the Frequency Range from 3 KHz to 300 GHz – Safety Code 6 (2015) of Health Canada (hereinafter "Safety Code 6"); and meets or exceeds standards for RF devices set out in the Radio Standards Specification 102, Radio Frequency (RF) Exposure Compliance of Radiocommunication Apparatus (All Frequency Bands), of Innovation, Science and Economic Development Canada. Note: The Canada Labour Code and the Canada Occupational Health and Safety Regulations do not apply to CAF members unless they have direct management or supervisory responsibilities in respect of DND employees.

Compliance and Requirements

5.2 The RFSP sets out: requirements for all DND and CAF organizations in which DND employees or CAF members work with or are near, hold or procure RF devices; requirements and procedures for the compliance evaluation of RF devices to be procured for use in DND and CAF organizations; procedures for the evaluation, prevention and control of hazards associated with RF devices that operate in the frequency range from 3 kHz to 300 GHz; requirements for the compliant use of RF devices during training and operations; emergency response procedures incidents involving RF radiation exposures; requirements and procedures for the notification, reporting and investigation of RF radiation exposures that constitute HOs; requirements for the medical assessment of overexposed personnel (see CF H Svcs Gp Instruction 4440-16, Physical Hazards Surveillance Program); requirements for DND employees and CAF members who use RF devices or work in an environment where RF radiation hazards may be present; and training and qualifications of RFSOs who are appointed to oversee the compliant use of RF devices and control RF radiation hazards.

Overriding RFS Standards

5.3 The Comd of a command or formation or the CO of a unit may override an RFS standard in Safety Code 6 if compliance would have a detrimental effect on activities in support of training and operations of the CAF. In such a case, the Comd or CO must: employ recognized risk management practices to determine an appropriate course of action; advise the RFSP manager within 24 hours; and be prepared to justify the decision in writing.

Compliance Evaluation Process

5.4 The process of evaluating RFSP compliance includes: L1 organizations conducting: evaluation of their RFSP; annual task forecasts and RFS evaluations of new RF devices; and regular RFS evaluations of existing RF devices; and the RFSP manager and RFS TA: reviewing the intended use of RF devices; performing hazard assessments on RF devices through data collection and analysis; providing recommendations and advice on measures required for achieving compliance; and reviewing preventive and protective measures implemented to control RF radiation hazards.

RFSOD Direction

5.5 The RFSOD provide detailed direction for: the safe use and control of RF devices; the procurement requirements for RF devices; the qualifications, training and evaluation of RFSOs; international obligations regarding the use of RF devices outside of Canada; requirements and procedures for conducting RFS surveys; procedures for RFSP evaluations; prevention, consequence control and emergency response for RF radiation exposure incidents; and requirements and procedures for the notification, reporting and investigation of RF radiation exposures that constitute HOs.

Medical Examinations

5.6 Requirements for medical examinations of DND employees and CAF members who use RF devices or work in an environment where RF radiation hazards may be present are outlined in CF H Svcs Gp Instruction 4440-16.

Materiel Acquisition and Support Process

5.7 RFSP requirements must be taken into account during the entire materiel acquisition and support process for an RF device.

5.8 No RF device is exempt from the compliance requirements of the RFSP. If a compliance evaluation has shown that an RF device presents an RF radiation hazard or is capable of causing injury during normal operation, appropriate physical and administrative hazard controls must be put in place in order to maintain exposure within the required limits.

5.9 In the case of a contract for a new or upgraded RF device: the contract must provide that: the contractor is required to provide HERP, HERF and HERO data as applicable; and the RF device is required to comply with Safety Code 6; and the RFS TA must verify compliance.

The RFSB

5.10 The RFSB: reports through the RFSP manager to ADM(Mat) on all RFS-related issues; may be represented on several safety committees; and consists of: a chairperson; and representatives from each of the following: L1 organizations holding or using RF devices; QETE; CF H Svcs Gp; Directorate of Ammunition and Explosives Regulation; and any other major stakeholders.

OHS Committees

5.11 In accordance with sections 134.1 and 135 of Part II of the Canada Labour Code, the National OHS Policy Committee and workplace OHS committees respectively must participate in the development, implementation and monitoring of OHS policies and programs. RFSP representatives are part of the National OHS Policy Committee, and local RFSP representatives are part of workplace OHS committees. The General Safety Program provides policy direction and guidance to the National OHS Policy Committee and workplace OHS committees.

6. Compliance and consequences

Compliance

6.1 DND employees and CAF members must comply with this DAOD. Should clarification of the policies or instructions set out in this DAOD be required, DND employees and CAF members may seek direction through their channel of communication or chain of command, as appropriate. Managers and military supervisors have the primary responsibility for and means of ensuring the compliance of their DND employees and CAF members with this DAOD.

Consequences of Non-Compliance

6.2 DND employees and CAF members are accountable to their respective managers and military supervisors for any failure to comply with the direction set out in this DAOD. Non-compliance with this DAOD may have consequences for both the DND and the CAF as institutions, and for DND employees and CAF members as individuals. Suspected non-compliance may be investigated. Managers and military supervisors must take or direct appropriate corrective measures if non-compliance with this DAOD has consequences for the DND or the CAF. The decision of an L1 or other senior official to take action or to intervene in a case of non-compliance, other than in respect of a decision under the Code of Service Discipline regarding a CAF member, will depend on the degree of risk resulting from the non-compliance and other circumstances of the case.

6.3 The nature and severity of the consequences resulting from non-compliance should be commensurate with the circumstances of the non-compliance and other relevant circumstances. Consequences of non-compliance may include one or more of the following: the ordering of the completion of appropriate learning, training or professional development; the entering of observations in individual performance evaluations; increased reporting and performance monitoring; the withdrawal of any authority provided under this DAOD to a DND employee or CAF member; the reporting of suspected offences to responsible law enforcement agencies; the application of specific consequences as set out in applicable laws, codes of conduct, and DND and CAF policies and instructions; other administrative action, including the imposition of disciplinary measures, for a DND employee; other administrative or disciplinary action, or both, for a CAF member; and the imposition of liability on the part of Her Majesty in right of Canada, DND employees and CAF members. Note: In respect of the compliance of DND employees, see the Treasury Board Framework for the Management of Compliance for additional information.

7. Responsibilities

Responsibility Table

7.1 The following table identifies the responsibilities associated with this DAOD:

COS(Mat) - providing guidance for the operation of the RFSP.

L1s advisors whose organizations hold or use RF devices - providing leadership and direction in respect of the operation of the RFSP; implementing the RFSP in their organizations; providing representatives to the RFSB; providing an annual forecast of survey tasks to ensure all existing RF devices are surveyed regularly according to the established levels of risk for each RF device; coordinating all RFS surveys and providing assistance and support to RFS personnel in performing RFS evaluations; reviewing and implementing recommendations to control RF radiation hazards; ensuring sufficient RFS expertise within their organizations to meet RFS requirements; and ensuring RFS control measures, including the appointment of RFSOs, are put in place, as required.

managers of a DND organization, Comds of a command or formation and COs of a unit that hold or use RF devices - ensuring DND employees and CAF members who use, hold, procure or work near RF devices are trained in accordance with relevant policies, orders, directives and guidance; ensuring RFS control measures are put in place through their own related orders, directives and instructions; and ensuring that all RF radiation exposures that constitute HOs are reported and investigated.

Surgeon General - providing advice to COS(Mat) as the RFS Champion on health risks associated with RF radiation exposures and suitability of protective measures; and providing occupational health support for RFSP development and sustainment, and the establishment of RFS standards for health effects associated with RF radiation exposures.

RFSP manager - developing policies, orders, directives and guidance for the RFSP; providing oversight of the RFSP; conducting RFS compliance monitoring; ensuring RF safety training is available to the DND and the CAF; responding to DND and CAF safety reporting requirements; as chairperson of the RFSB; maintaining awareness of current developments in national and international RFS standards, policies and procedures; and making recommendations on: research or trials to develop or improve the RFSP; and appropriate control of RF radiation exposures from RF devices; and addressing other RFS-related issues, as required.

RFS TA - providing RFS technical delivery and expertise, engineering support and guidance to L1 organizations; providing advice to L1s on the identification, analysis and control of RF radiation hazards; interpreting RFS standards; coordinating with Health Canada and Innovation, Science and Economic Development Canada on all matters pertaining to the interpretation and application of Safety Code 6; and specifying RFS control measures.

RFSOs - providing advice on the implementation and management of the RFSP at their DND organizations, units and other elements; implementing and coordinating RFS control measures on behalf of their respective managers, Comds and COs; ensuring all RF devices within their organizations are properly installed, maintained, surveyed and operated in accordance with the RFSP and relevant policies, orders, directives and guidance; and ensuring all DND employees and CAF members who use RF devices, or work in an environment where RF radiation hazards may be present, receive RFS training.

life cycle materiel managers, procurement officers and project managers - ensuring all RF devices under their control have been evaluated to establish the extent and type of their RF radiation hazards; ensuring, in the case of any RF device that presents an RF radiation hazard or is capable of causing injury during normal operation, that appropriate physical and administrative hazard controls are put in place in order to maintain exposures within the required limits; ensuring that DND contractors provide HERP, HERF and HERO data on new, modified or upgraded RF devices as part of contract deliverables; ensuring that contracts comply with Safety Code 6; and ensuring that RFS compliance evaluations and RFS surveys are conducted for relevant HERP, HERF and HERO requirements, as part of first article testing for any new, modified or upgraded RF device, by requesting that the RFS TA either: conduct both the evaluation and the survey; or conduct the evaluation and use a contractor to conduct the survey under the following conditions: a detailed test plan is submitted to the RFS TA at least 60 days prior to starting the survey; RFS TA test protocols are used during the survey; and the RFS TA maintains oversight and validates all results of the survey. Note: Details of the requirements for using contractors are set out in C-55-040-001/TS-002.

DND employees and CAF members using RF devices - taking required RFS training; complying with RFS control measures set out in relevant policies, orders, directives and guidance; and reporting all RF radiation exposures that constitute HOs.

8. References

Acts, Regulations, Central Agency Policies and Policy DAOD:

Explosives Act

Canada Labour Code, Part II, Occupational Health and Safety

Canada Occupational Health and Safety Regulations

Framework for the Management of Compliance, Treasury Board

Limits of Human Exposure to Radiofrequency Electromagnetic Fields in the Frequency Range from 3 KHz to 300 GHz – Safety Code 6 (2015), Health Canada

DAOD 3026-0, Radio Frequency Safety

Other References:

DOAD 1000-8, Policy Framework for Safety and Security Management

DAOD 2007-0, Safety

DAOD 3000-0, Materiel Acquisition and Support

DAOD 3002-0, Ammunition and Explosives

DAOD 3002-3, Ammunition and Explosives Safety Program

DAOD 4002-1, Nuclear and Ionizing Radiation Safety

DAOD 6002-4, Radio Frequency Spectrum Management

CANFORGEN 122/07, Applicability of the Canada Labour Code, Part II and the General Safety Program to CF Members and DND Employees

A-GG-040-001/AG-001, General Safety Program, Volume 1, Policy and Program

A-GG-040-006/AG-001, DND Explosives Safety Program

C-09-005-009/TS-001, Ammunition and Explosives Safety Manual, Volume 9, Hazards of Electromagnetic Radiation to Ordnance (HERO)

C-55-040-001/TS-002, Radio Frequency Safety Standards and Requirements

CF Health Services Group Instruction 4440-16, Physical Hazards Surveillance Program

Radio Standards Specification 102, Radio Frequency (RF) Exposure Compliance of Radiocommunication Apparatus (All Frequency Bands), Innovation, Science and Economic Development Canada

GL-01, Guidelines for the Measurement of Radio Frequency Fields at Frequencies from 3 KHz to 300 GHz, Innovation, Science and Economic Development Canada

STANAG 2345, Military Workplaces – Force Health Protection Regarding Personnel Exposure to Electric, Magnetic, and Electromagnetic Fields, 0 Hz to 300 GHz